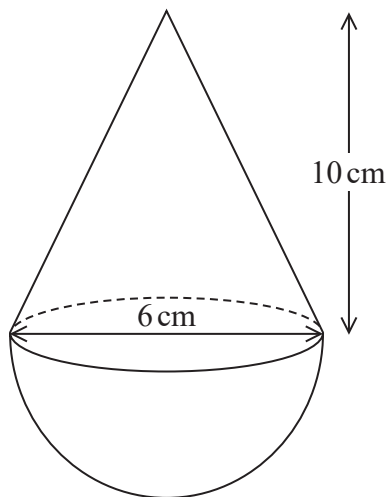
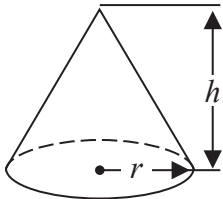


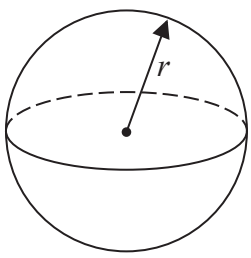
- 15 The diagram shows a solid shape.
The shape is a cone on top of a hemisphere.



Volume of a cone = $\frac{1}{3} \pi r^2 h$

A diagram of a cone. A horizontal line from the center of the base to the edge is labeled 'r'. A vertical line from the apex to the center of the base is labeled 'h'.

Volume of a sphere = $\frac{4}{3} \pi r^3$

A diagram of a sphere. A line from the center to the outer surface is labeled 'r'.

The height of the cone is 10 cm.
The base of the cone has a diameter of 6 cm.
The hemisphere has a diameter of 6 cm.

The total volume of the shape is $k\pi \text{ cm}^3$, where k is an integer.

Work out the value of k .

$k = \dots\dots\dots$